

A Robust Evaluation Framework for Lightweight Vision Transformers for Multi-Class Deepfake Detection on Resource Constrained Devices

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Abstract— The rapid advancement of Generative Artificial Intelligence has produced highly realistic face forgeries that pose significant challenges for digital forensics, media authentication, and online trust. Although Vision Transformers have demonstrated strong performance for deepfake detection, their computational complexity limits deployment on resource constrained devices, and most existing approaches focus only on binary classification. This paper presents an end-to-end framework for robust multi-class face forgery detection using lightweight Vision Transformer architectures. The proposed framework integrates balanced dataset construction, preprocessing, transfer learning, class aware optimization through weighted loss, Mixup, CutMix, and label smoothing, followed by comprehensive evaluation of detection accuracy, computational efficiency, confidence calibration, and cross dataset generalization. Four lightweight Vision Transformers, MobileViT, MobileViTv2, EdgeNeXt, and EfficientViT, are evaluated on the DF40 benchmark and validated on the FaceForensics++ and Deepfake Detection Localization datasets. Experimental results show that MobileViTv2 achieves the best overall balance between accuracy and efficiency, attaining 96% multi-class accuracy with only 2.56 ms inference latency and 4.4 million parameters. The proposed framework provides a practical benchmark for developing robust, efficient, and trustworthy deepfake detection systems for edge devices.

Keywords—Deepfake detection, Face Forgery detection, Vision Transformers, Lightweight Vision Transformers, Edge AI

I. INTRODUCTION

The rapid proliferation of highly realistic face manipulation techniques has become one of the most significant challenges in digital forensics. Recent advances in Deep Learning (DL) and Generative Artificial Intelligence (GenAI), particularly Generative Adversarial Networks (GANs) and Diffusion Models (DMs), are capable of synthesizing facial images and videos that are nearly indistinguishable from authentic media. While these technologies have enabled numerous beneficial applications in entertainment, education, and content generation, they have also facilitated malicious activities including identity theft, misinformation campaigns, financial fraud, political disinformation, and sextortion. As manipulated media becomes increasingly accessible and convincing, manual inspection is no longer sufficient, creating an urgent need for automated deepfake detection systems that are accurate, reliable, efficient, and trustworthy.

Early deep-fake detection methods were primarily based on Convolutional Neural Networks (CNNs), which demonstrated strong performance by learning discriminative local visual features. However, modern face manipulation techniques often introduce subtle inconsistencies that are distributed across multiple facial regions rather than being confined to localized artifacts. Consequently, CNNs may fail to capture these long-range relationships because of their inherently limited receptive fields. Vision Transformers (ViTs) have recently emerged as a promising alternative by employing self-attention mechanisms that model global contextual dependencies across an image. This ability to jointly analyze distant facial regions enables ViTs to better identify manipulation artifacts that are difficult to detect using conventional convolutional architectures.

Despite their superior representational capabilities, conventional Vision Transformers remain computationally expensive due to the quadratic complexity of the self-attention operation with respect to the number of input tokens. Such computational and memory requirements limit their deployment on resource-constrained platforms, including smartphones, embedded systems, and edge devices where real-time inference is often required. To overcome these limitations, several lightweight Vision Transformer architectures, including MobileViT, MobileViTv2, EdgeNeXt, and EfficientViT, have been proposed. These architectures employ efficient design strategies such as separable self-attention, inverted residual blocks, and cascaded group attention to significantly reduce computational overhead while preserving the global modeling capabilities of transformer-based networks. Consequently, lightweight ViTs have become attractive candidates for real-time face forgery detection in practical edge-computing environments. Although lightweight Vision Transformers have demonstrated encouraging performance on benchmark datasets, several important challenges remain unresolved. *First*, most existing studies evaluate models only on the datasets used for training, where high accuracy may simply reflect the learning of dataset-specific artifacts rather than genuine forensic traces. As a result, detection performance frequently deteriorates when models are evaluated on previously unseen datasets or new manipulation techniques, exposing a significant cross-dataset generalization gap.

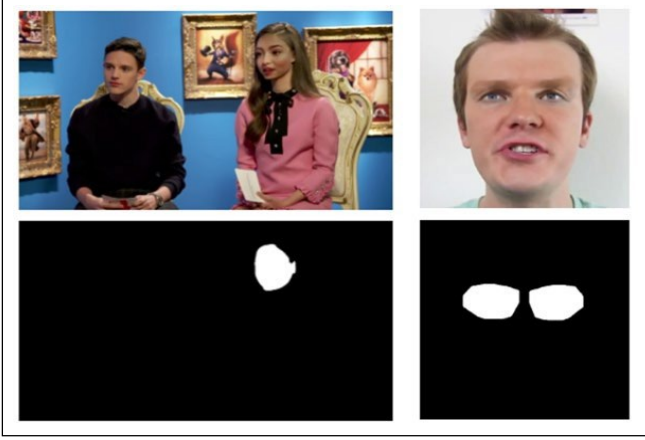


Fig. 1. Samples from DDL with their ground truth masks

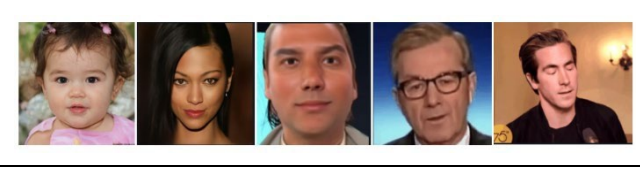


Fig. 2. Samples from DF40

Table 1 Evolution of the Proposed Lightweight Vision Transformer Training Framework

Component	Phase 1 (Previous Work) [11]	Phase 2 (Current Work)
Task	Binary	Five-Class
Training	Linear Probing + Layer-wise FT	Same
Loss	Cross Entropy	Weighted Cross Entropy
Mixup	No	Yes
CutMix	No	Yes
Label smoothing	No	Yes
Objective	Baseline	Robustness & Generalization

Second, the majority of existing work formulates deepfake detection as a binary classification problem that distinguishes only between authentic and manipulated images, providing limited understanding of a model's ability to recognize different categories of facial forgeries. These limitations reduce the reliability and trustworthiness of current deepfake detection systems, particularly in forensic and legal applications where model decisions must be transparent and defensible.

To address these challenges and building upon our previous work [11] of evaluating lightweight vision transfers for binary classification only, this paper presents an end-to-end framework for robust multi-class face forgery detection using lightweight Vision Transformer architectures. Table 1 summarizes the main evolution from the previous binary-classification framework to the current multi-class robustness and optimization framework. Unlike conventional approaches that primarily focus on binary classification, the proposed framework supports both binary detection (Real versus Fake) and multi-class forgery classification involving Face Reenactment (FR), Face Swapping (FS), Entire Face Synthesis (EFS), and Face Editing (FE). The framework further incorporates balanced dataset construction, preprocessing and data integrity verification, class-aware learning through weighted loss functions, advanced regularization techniques, and comprehensive performance

evaluation. In addition to measuring predictive performance, the proposed framework explicitly investigates cross-dataset generalization. Collectively, these components provide a comprehensive assessment of lightweight Vision Transformer architectures for practical deepfake detection on resource-constrained devices.

II. RESEARCH CONTRIBUTIONS

The contributions of this paper are listed as follows:

- A. End-to-End Unified Framework:** We propose a unified four-stage framework consisting of data preparation and balancing, preprocessing and data integrity verification, model training and optimization and performance. A high-level overview of the framework for consistency is shown in Fig. 3. The framework is specifically designed to improve robustness, generalization, and deployment efficiency for lightweight Vision Transformer-based deepfake detection.
- B. Comprehensive Evaluation of Lightweight Vision Transformers:** We perform an extensive comparative evaluation of MobileViT, MobileViTv2, EdgeNeXt, and EfficientViT across both binary and multi-class face forgery detection tasks, analyzing their detection accuracy, inference latency, parameter efficiency, and model complexity.
- C. Extension from Binary to Multi-Class Forgery Detection:** Building upon our previous work, i.e., binary (conventional real-versus-fake classification), we investigate the more challenging task of distinguishing among four major forgery categories: Face Reenactment (FR), Face Swapping (FS), Entire Face Synthesis (EFS), and Face Editing (FE).
- D. Improving Robustness Through Regularization and Class-Aware Learning:** To improve model reliability beyond the source dataset, the proposed framework incorporates balanced dataset construction, weighted loss functions, Mixup and CutMix augmentation, and optimized fine-tuning strategies. The resulting models are evaluated on external datasets to assess their ability to generalize previously unseen forgery distributions.

The experimental results demonstrate the trade-offs between accuracy, efficiency and robustness providing practical insights into the suitability of lightweight Vision Transformer architectures for real-world edge-based deepfake detection systems.

III. RELATED WORK

A. Evolution of Deepfake Benchmarks

The evolution in deepfake detection has been tied to the evolution of benchmark datasets. First-generation sets, such as FaceForensics++ (FF++, 2019) [5], created a foundation by using classical computer graphics-based methods like Face2Face and NeuralTextures. However, recent literature indicates that FF++ is being considered “outdated” as it contains limited forgery methods that do not represent the level of modern neural synthesis [6]. To address the

limitations, the DF40 dataset (2024) [7] introduces a large-scale benchmark, encompassing 40 distinct techniques including modern synthesis methods. Furthermore, the DDL dataset (2025) [8] has emerged as an important benchmark for interpretability, providing over 1.5 million images with pixel-level ground-truth masks to objectively evaluate model localization. Fig. 1 shows some high quality samples of the DDL with ground truth mask and Fig. 2 of DF40, illustrating how indistinguishable those fake samples can be in comparison to authentic images.

B. Lightweight Architectures for Edge-Device Forensics

While large-scale architectures like XceptionNet (Convolutional Neural Network) [5] and Swin-B (Vision Transformer) [9] achieve near perfect accuracy on sets like FF++, they have high parameters count which makes them incompatible for deployment on constrained devices. These results accelerated research into using lightweight Vision Transformers (ViTs), since they combine the global receptive fields of transformers with the efficiency and local feature extraction of convolutional layers. While MobileViT (2022) used a methodology to combine both Transformer and Neural Network layers for edge-device model development [1], they fall short in addressing the quadratic complexity of self-attention. MobileViTv2 (2022) addressed the complexity issue by introducing separable self-attention, reducing the computational requirements from quadratic to linear (with respect to the number of tokens) [2]. Similarly, EdgeNeXt (2022) utilized Split Depth-wise Transpose Attention (SDTA) to optimize global context encoding [3]. Moreover, EfficientNet (2023) pushed the efficiency limits by using cascaded group attention to minimize memory-bounded operations [4]. This research makes use of these innovations to determine the optimal balance for real-world deepfake detection.

IV. PROPOSED END-TO-END DETECTION FRAMEWORK

A. Framework Motivation

This paper builds upon our previous research on lightweight Vision Transformer (ViT) architectures for face forgery detection. In our preliminary study, we established a baseline using ImageNet-pretrained lightweight ViT backbones through a Linear Probing strategy, where the feature extraction layers were frozen to evaluate the effectiveness of pretrained representations for forgery detection. This was followed by a Layer-wise Fine-tuning approach, where the final transformer layers and classification head were unfrozen to adapt the models to forensic-specific features while preserving the learned low-level visual representations. Using this training strategy, MobileViTv2 achieved 98% accuracy for binary classification and 96% accuracy for multi-class classification on the DF40 benchmark, demonstrating the effectiveness of lightweight ViTs for efficient face forgery detection. The primary objective of this paper is to extend our previous framework beyond conventional binary deepfake detection toward a more comprehensive and robust face forgery detection pipeline [15]. In particular, this paper advances the framework by supporting fine-grained multi-class forgery classification involving Face Reenactment (FR), Face Swapping (FS), Entire

Face Synthesis (EFS), and Face Editing (FE). Furthermore, the proposed framework addresses the cross-dataset generalization gap observed in our previous work through balanced dataset construction, optimized training strategies, and enhanced regularization. These enhancements improve the robustness and generalization capability of lightweight Vision Transformer models, enabling more reliable face forgery detection across diverse datasets and manipulation techniques while maintaining computational efficiency for deployment on resource-constrained devices.

B. Framework Overview

Fig 2., presents representative samples from the DF40 benchmark dataset used in this paper. The dataset comprises authentic facial images together with manipulated images generated using forty state-of-the-art face forgery techniques, spanning four major forgery categories: Face Reenactment (FR), Face Swapping (FS), Entire Face Synthesis (EFS), and Face Editing (FE). The subtle visual differences between authentic and manipulated samples highlight the complexity of modern face forgery detection and the need for robust learning frameworks capable of capturing both local forensic artifacts and global contextual information.

Building upon the motivation described in the previous subsection, Fig. 4 illustrates the proposed end-to-end detection framework. The framework is organized into four sequential stages: (1) dataset preparation and class balancing, (2) preprocessing and data integrity verification, (3) model training and optimization, and (4) performance evaluation. Each stage addresses a specific component of the face forgery detection pipeline while collectively improving model robustness, computational efficiency, and cross-dataset generalization.

The framework begins by constructing a balanced training dataset through the integration of authentic facial images from multiple public datasets with manipulated samples from DF40. The collected data are then preprocessed and verified to ensure consistency and data integrity before being supplied to the learning models. Lightweight Vision Transformer architectures are subsequently trained using transfer learning, class-aware optimization, and advanced regularization techniques to learn discriminative forensic representations while maintaining computational efficiency for deployment on resource-constrained devices. Finally, the trained models are comprehensively evaluated using quantitative performance metrics, cross-dataset validation, and analysis to assess their effectiveness, robustness, and generalization capability. The following subsections describe each stage of the proposed framework in detail.

1) Data Set Preparation and Class Balancing

The first stage of the proposed framework focuses on constructing a balanced and representative dataset to facilitate robust model training and improve generalization. The DF40 benchmark serves as the primary dataset for model development, containing more than one million manipulated facial images generated using forty state-of-the-art face forgery techniques. Unlike earlier benchmarks, DF40 supports both binary and multi-class forgery detection and

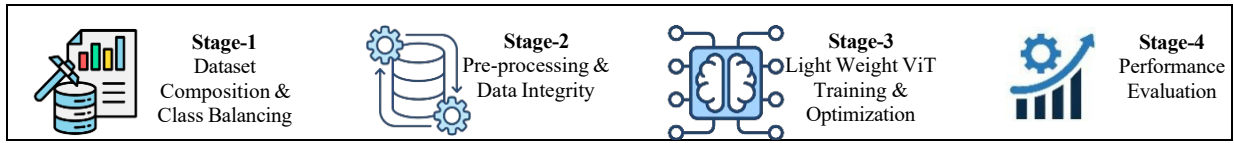


Fig. 3. Samples from DF40

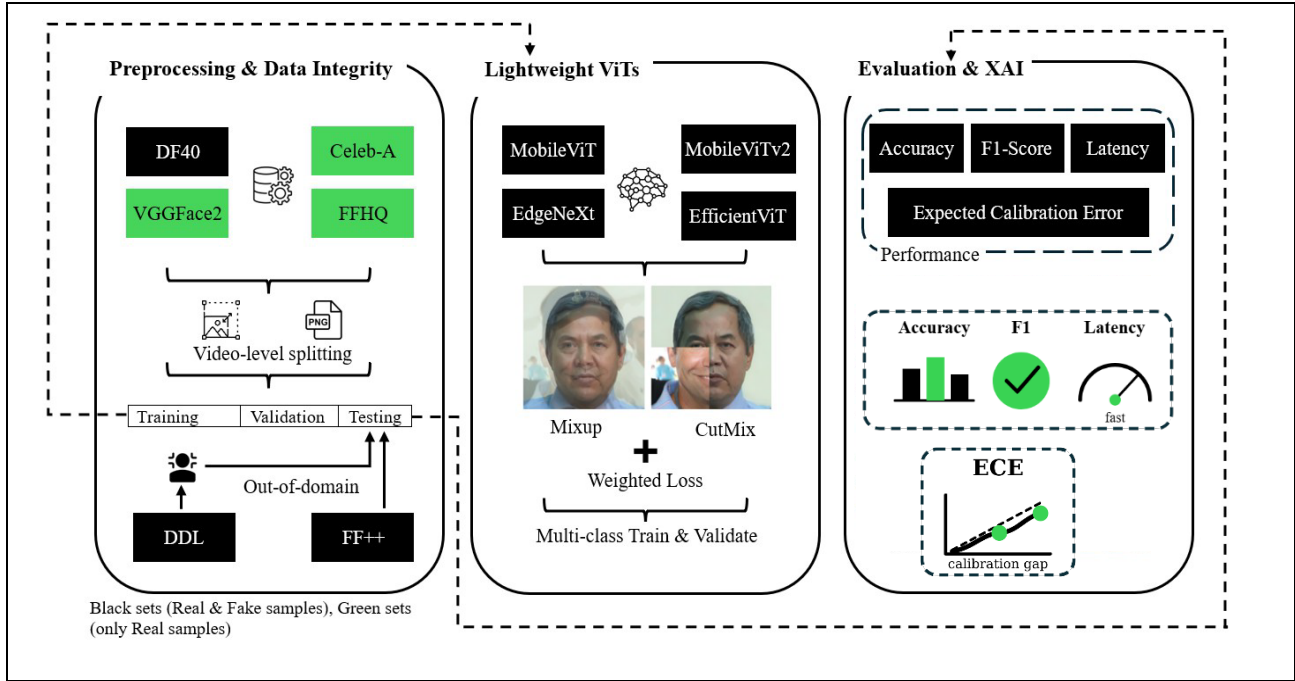


Fig. 4. The proposed end-to-end framework for deepfake detection

provides a diverse collection of manipulation methods representative of modern deepfake generation techniques. One of the primary challenges associated with DF40 is the severe imbalance between authentic and manipulated images, with an approximate Real-to-Fake ratio of 1:20. Such imbalance can bias the learning process toward the majority class, resulting in reduced sensitivity toward the authentic facial images and limiting the model's ability to generalize. To mitigate this issue, additional authentic facial images were incorporated from three publicly available datasets: Celeb-A [11] (200,000 images), VGGFace2 [12] (180,000 images) and FFHQ [10] (52,000 images), increasing the diversity of authentic facial characteristics while reducing the overall class imbalance to approximately 1:3. The resulting dataset provides a more representative distribution of authentic and manipulated samples, enabling the lightweight Vision Transformer models to learn more discriminative and robust facial representations.

To further assess the robustness of the proposed framework, two independent benchmark datasets were employed during evaluation. FaceForensics++ (FF++) was used to measure cross-dataset generalization by evaluating model performance on unseen manipulation techniques, while the Deepfake Detection Localization (DDL) benchmark was utilized to evaluate robustness. These external datasets provide a comprehensive assessment of the proposed framework beyond the source training distribution.

2) Preprocessing and Data Integrity Verification

The second stage of the proposed framework focuses on preprocessing and data integrity verification to ensure that the training data are consistent, unbiased, and representative of real-world deployment conditions. The preprocessing pipeline was designed to mitigate common challenges in deepfake detection, including data leakage, source-domain bias, dataset inconsistencies, and image quality variations. The major preprocessing operations are summarized below.

(a). **Video-Level Dataset Splitting:** To ensure scientific validity and eliminate data leakage, a video-level splitting strategy was adopted. Rather than randomly partitioning individual frames, all frames extracted from the same source video were assigned exclusively to either the training, validation, or testing subset. This prevents highly correlated samples from appearing across multiple splits, avoiding artificially inflated performance estimates and providing a more realistic assessment of model generalization.

(b). **Image Standardization:** Since the collected datasets contain images with varying resolutions, aspect ratios, and file formats, all samples were standardized prior to training. Images were resized to match the native input resolution of the corresponding lightweight Vision Transformer architecture (224×224 or 256×256 pixels) and converted to PNG format. Standardizing the image format minimizes variations caused by compression artifacts and encoding differences, allowing the models to focus on intrinsic facial manipulation features rather than dataset-specific image characteristics.

(c). **Data Augmentation:** To improve model robustness and reduce overfitting, online data augmentation was performed following the recommendations provided in the original DF40 benchmark. Each operation was applied with a probability of 50%, including random horizontal flipping, random rotations between -10° and $+10^\circ$, Gaussian blurring using kernel sizes ranging from 3 to 7, and color jittering with brightness and contrast variations of 0.1. Additionally, random JPEG compression with quality factors between 40 and 100 was introduced to simulate real-world image degradation, encouraging the models to learn manipulation features that remain invariant to common compression artifacts and digital noise.

(d). **Face Extraction and Label Assignment:** This preprocessing step was applied exclusively to the DDL benchmark. Unlike DF40, which primarily contains single-face images, DDL frequently includes multiple faces within a single image. Since this study focuses on single-face forgery detection, Google's BlazeFace detector was employed to identify and crop individual facial regions. The cropped faces were automatically labeled using the corresponding ground-truth manipulation masks, where a crop was assigned the Fake label if the associated mask contained manipulated pixels; otherwise, it was labeled Real. This automated procedure ensured consistent annotation while preserving compatibility with the proposed single-face detection framework.

3) Model Training and Optimization

The third stage of the proposed framework focuses on training and optimizing lightweight Vision Transformer (ViT) architectures for robust face forgery detection while maintaining computational efficiency suitable for resource-constrained devices. Since each architecture is available in multiple configurations, model variants with comparable numbers of trainable parameters were selected to ensure a fair comparison. Specifically, MobileViT-S (4.94 million parameters), MobileViTv2 (4.39 million parameters), EdgeNeXt-S (7.44 million parameters), and EfficientViT-M3 (6.85 million parameters) were evaluated. All models were initialized using ImageNet-pretrained weights to leverage rich visual representations learned from large-scale natural image datasets. The classification head was initialized using Xavier uniform initialization with biases set to zero, while the AdamW optimizer was employed throughout all experiments due to its stability and effectiveness for transformer-based architecture.

To systematically develop and evaluate the lightweight Vision Transformer models, the proposed training methodology was divided into two sequential phases. The first phase established a robust binary classification baseline, while the second phase extended the framework to fine-grained multi-class face forgery classification through additional optimization and regularization strategies.

Phase 1: Binary and Multi-class Baseline and Model Stabilization (Previous Work): The first phase focused on the conventional binary face forgery detection task (Real vs. Fake) to establish a performance baseline. Transfer learning was performed using a two-stage strategy consisting of Linear

Probing followed by Layer-wise Fine-tuning. Linear Probing evaluated the effectiveness of ImageNet-pretrained feature representations by freezing the backbone and training only the classification head. Subsequently, the final transformer layers and classification head were fine-tuned to adapt the pretrained models to the forensic domain while preserving robust low-level visual representations. This strategy demonstrated the effectiveness of lightweight Vision Transformers on the large-scale DF40 benchmark and established the baseline for the current study.

Phase 2: Multi-Class Robustness and Optimization (Current Work): Building upon the baseline established in Phase 1, the second phase extends the problem from binary classification to a five-class taxonomy consisting of Real, Face Reenactment (FR), Face Swapping (FS), Entire Face Synthesis (EFS), and Face Editing (FE). The increased complexity of distinguishing multiple forgery categories, together with the significant intra-class variation among manipulated images, required additional optimization and regularization strategies to improve robustness and generalization. Accordingly, the Phase 2 training pipeline incorporates three key optimization strategies designed to improve learning stability and model robustness: inverse class frequency weighting, aggressive regularization (Mixup and CutMix), and label smoothing.

- **Inverse Class Frequency Weighting:** To compensate for the under-representation of minority classes, particularly the Face Editing (FE) category, weighted cross-entropy loss was employed during optimization. Class weights were computed using the inverse class frequency as shown in equation (i).

$$W_c = \frac{N}{C \times N_c} \quad \dots\dots(i)$$

where (N) denotes the total number of training samples, (C) represents the total number of classes, and (N_c) is the number of samples belonging to class (c). This weighting scheme assigns greater importance to minority classes, encouraging balanced learning across all forgery categories.

- **Aggressive Regularization (Mixup and CutMix):** To improve model robustness and reduce overfitting, Mixup ($\alpha = 0.8$) and CutMix ($\alpha=1.0$) augmentations were incorporated during training. These techniques generate synthetic training samples by combining images and their corresponding labels, encouraging smoother
- **Label Smoothing:** A label smoothing factor of 0.1 was applied during training to soften the target probability distribution. This regularization strategy discourages overconfident predictions, improves model calibration, and further enhances generalization performance.

4) Evaluation Metrics

To comprehensively evaluate the effectiveness of the proposed framework for both binary and multi-class face forgery detection, the trained models were assessed from three complementary perspectives: *classification performance*, *prediction reliability*, and *computational efficiency*. Together, these evaluation criteria provide a balanced assessment of

model accuracy, robustness, confidence calibration, and deployment suitability for real-world digital forensic applications.

(a). *Classification Performance* -Standard classification metrics were employed to evaluate detection performance for both the binary and multi-class classification tasks. Overall Accuracy measures the proportion of correctly classified samples over the entire testing set. Since the multi-class dataset contains varying class distributions, the Macro F1-score was additionally adopted to provide an unbiased evaluation by assigning equal importance to each forgery category. The F1-score for class (c) is defined as the harmonic mean of precision and recall as listed in equation (ii). Whereas, Overall accuracy, shown in equation (iii), is the ratio of correct predictions to the number of available samples. Macro F1-score, i.e., equation (iv), is the unweighted mean of all the C classes F1-scores.

$$F1 - score(c) = \frac{2 * Precision\ c * Recall\ c}{(Precision\ c + Recall\ c)} \dots (ii)$$

$$Overall\ accuracy = \frac{Correct\ prediction}{Testing\ sample\ size} \dots (iii)$$

$$Macro\ F1 = \frac{1}{C} \sum_{i=1}^C F1_i \dots (iv)$$

While the above metrics quantify classification performance, they do not indicate whether the predicted confidence scores are reliable. Therefore, an additional calibration metric was incorporated to evaluate prediction confidence.

5) Prediction Reliability:

Beyond classification accuracy, the reliability of the predicted confidence scores was assessed using the Expected Calibration Error (ECE). In digital forensic applications, highly confident yet incorrect predictions can significantly reduce trust in automated detection systems. ECE in equation (v) quantifies the agreement between predicted confidence and observed accuracy by partitioning predictions into (K) confidence bins and computing the weighted difference between the average confidence and classification accuracy within each bin.

$$ECE = \sum_{k=1}^K \frac{|B_k|}{n} |acc(B_k) - conf(B_k)| \quad (v)$$

where (B_k) denotes the set of samples belonging to confidence bin (k), (n) is the total number of predictions, ($acc(B_k)$) represents the classification accuracy within the bin, and ($conf(B_k)$) denotes the corresponding average confidence score. While classification accuracy and calibration assess predictive performance, they do not capture the computational efficiency required for deployment on resource-constrained devices. Consequently, the proposed framework was also evaluated using standard computational complexity metrics.

6) Computational Benchmarking:

Theoretical complexity was quantified using Multiply-Accumulate Operations (MAdds) and total parameter counts. MAdds measure the number of arithmetic operations per

forward pass, where each multiplication and its associated addition are counted as one. The number of trainable parameters is an indication of the model size; theoretical complexity alone is insufficient to determine real-world efficiency due to architectural bottlenecks and memory access overheads. Hence, we measure the model latency to assess practical efficiency. All the measurements were conducted on consistent hardware (NVIDIA H200) in a stable state (after warm-up iterations). Since the DDL and FaceForensics++ benchmarks contain only binary labels (Real and Fake), the outputs of the five-class classifier were converted into binary predictions using an aggregate fake probability. The predicted probabilities for Face Reenactment (FR), Face Swapping (FS), Entire Face Synthesis (EFS), and Face Editing (FE) were summed to form a single Fake probability, which was then compared against the Real probability for evaluation on the binary benchmark datasets.

Table 2 Model Accuracy & F1-scores

	Model	Accuracy	Macro Fake F1	Real F1
Previous Results	MobileViT	66%	59%	15%
	MobileViTv2	96%	90%	62%
	EdgeNeXt	95%	90%	60%
	EfficientViT	89%	80%	31%
Weighted Loss	MobileViT	96%	91%	68%
	MobileViTv2	96%	91%	66%
	EdgeNeXt	95%	87%	49%
	EfficientViT	91%	80%	26%
Reg + Weighted Loss	MobileViT	96%	91%	67%
	MobileViTv2	96%	92%	68%
	EdgeNeXt	89%	88%	23%
	EfficientViT	92%	81%	28%

V. RESULTS

This section presents a comprehensive evaluation of the proposed lightweight Vision Transformer framework for multi-class face forgery detection. Following the methodology described in the previous section, the experiments are organized into three complementary analyses. First, we investigate the impact of the proposed optimization strategies, including inverse class frequency weighting and aggressive regularization, on classification performance. Next, we evaluate the computational efficiency and prediction reliability of the optimized models to assess their suitability for deployment on resource-constrained devices. Finally, we examine the cross-dataset generalization capability of the proposed framework using two independent benchmark datasets, FaceForensics++ (FF++) and the Deepfake Detection Localization (DDL) benchmark.

1) *Impact of Class Weighting and Regularization on Performance:* To evaluate the effectiveness of the proposed optimization strategies, we first analyze the individual and combined impact of Inverse Class Frequency Weighting and aggressive regularization techniques (Mixup, CutMix, and Label Smoothing). Table 2 summarizes the resulting Overall Accuracy, Macro F1-score, and Real-class F1-score for each

lightweight Vision Transformer architecture. By comparing these results with the baseline established in our previous work, we quantify the contribution of each optimization strategy.

a) Class Weighting Observations: The introduction of inverse class frequency weighting produced noticeably different responses across the four architectures, highlighting that the effectiveness of weighted learning is architecture dependent.

- 1) **MobileViT** exhibited the largest improvement, transitioning from the weakest baseline performer (66% accuracy) to one of the best-performing models with 96% accuracy and a Real F1-score of 68%. This substantial gain suggests that MobileViT is highly sensitive to class imbalance and benefits significantly from class-aware optimization.
- 2) **MobileViTv2** maintained its position as the most stable architecture, preserving its baseline accuracy of 96% while achieving modest improvements in both Macro F1-score and Real F1-score. These results indicate that MobileViTv2 is inherently more robust to class imbalance.
- 3) In contrast, **EdgeNeXt** and **EfficientViT** demonstrated an unexpected response. Although their overall accuracies remained relatively stable, the Real-class F1-score decreased from 60% to 49% for EdgeNeXt and from 31% to 26% for EfficientViT. This behavior suggests that weighted loss may have destabilized the optimization process for these architectures. While class weighting improved minority-class representation, additional regularization was introduced to investigate whether model robustness and generalization could be further enhanced.

b) Aggressive Regularization Observations: Incorporating Mixup, CutMix, and Label Smoothing together with weighted loss further differentiated the behavior of the evaluated architectures.

- 1) **MobileViTv2** remained the most consistent performer, achieving the highest overall balance between accuracy and F1 scores while demonstrating small but consistent performance improvements.
- 2) **MobileViT** largely preserved the gains obtained through class weighting, indicating that the additional regularization neither degraded nor substantially improved its already optimized performance.
- 3) **EdgeNeXt** continued to exhibit counter-intuitive behavior, with its Real F1-score further decreasing to 23%, suggesting that stronger regularization adversely affected its ability to recognize authentic facial images.
- 4) **EfficientViT** showed only marginal improvements, indicating that aggressive regularization had a limited influence on its overall classification performance.

Table 2 reveals that no single optimization strategy consistently benefits all lightweight Vision Transformer architectures. MobileViT and MobileViTv2 exhibit significant performance improvements, whereas EdgeNeXt and EfficientViT show limited or even adverse responses,

highlighting the importance of architecture-specific optimization for multi-class face forgery detection.

Table 3 Efficiency and Reliability Metric

Model	Latency (ms)	MAdds (G)	Parameters (M)	ECE (Weighted Loss)	ECE (Weighted Loss + Reg)
MobileViT	3.53	2.1	4.9	0.64	0.47
MobileViTv2	2.56	1.9	4.4	0.66	0.47
EdgeNeXt	2.95	1.6	7.4	0.64	0.46
EfficientViT	5.60	0.3	6.6	0.61	0.45

2) Computational Benchmarking and Calibration Analysis

While classification performance is essential, practical deployment on resource-constrained devices also requires computational efficiency and reliable confidence estimation. Therefore, we next evaluate the computational characteristics and prediction calibration of the optimized models. Table 3 summarizes inference latency, computational complexity (MAdds), model size, and Expected Calibration Error (ECE). Based on these results, two complementary aspects are analyzed: (1) computational efficiency, which assesses the suitability of each architecture for real-time deployment, and (2) prediction calibration, which evaluates the reliability of the confidence scores produced by the models. The key observations from each analysis are summarized below.

- **Efficiency Observations:** Although EfficientViT exhibits the lowest theoretical computational complexity at only 0.3 G MAdds, it surprisingly records the highest inference latency (5.60 ms). This discrepancy suggests that theoretical complexity alone does not accurately predict practical runtime performance, likely due to architectural overhead associated with its specialized attention mechanisms. In contrast, MobileViTv2 achieved the lowest latency (2.56 ms) while maintaining the smallest parameter count (4.4 million), making it the most suitable architecture for real-time deployment. While computational efficiency determines deployment feasibility, reliable confidence estimation is equally important in digital forensic applications where incorrect high-confidence predictions can significantly reduce user trust.

- **Calibration Observations:** The introduction of aggressive regularization consistently reduced Expected Calibration Error across all evaluated architectures. ECE values decreased from approximately 0.64–0.66 to 0.45–0.47, demonstrating that Mixup, CutMix, and Label Smoothing effectively improve confidence calibration by discouraging overconfident predictions. Despite this improvement, calibration remains a significant challenge for multi-class face forgery detection. ECE values close to 0.45 are substantially higher than those observed in our previous binary classification framework (approximately 0.05), indicating that although the proposed models become better calibrated, their predicted confidence scores still deviate considerably from the true probability of correctness. This behavior likely reflects the increased complexity and intra-class variability associated with distinguishing multiple forgery categories.

3) Generalization and Out-of-Domain Robustness

While the previous sections demonstrated that the proposed optimization strategies improve in-domain classification performance, computational efficiency, and prediction reliability, it is equally important to determine whether these improvements generalize to previously unseen datasets. To evaluate the robustness of the optimized models, the best-performing configurations (Weighted Loss + Regularization) were evaluated on the FaceForensics++ (FF++) and Deepfake Detection Localization (DDL) benchmarks. Since both datasets provide only binary labels (Real and Fake), whereas the proposed models were trained using a five-class objective, the Aggregate Fake Score strategy described in Section IV was employed to convert the multi-class predictions into binary decisions. The resulting out-of-domain performance is summarized in Table 4.

Table 4 Out-of-domain benchmarking results

Model	FF++ Accuracy	Fake F1	Real F1	DDL Accuracy	Fake F1	Real F1
MobileViT	66%	85%	34%	58%	71%	53%
MobileViTv2	63%	82%	32%	56%	67%	46%
EdgeNeXt	71%	73%	42%	60%	58%	55%
EfficientViT	75%	76%	43%	64%	60%	56%

FF++ Evaluation: Although the evaluated models achieved relatively high overall accuracies ranging from 63% to 75%, these results become less encouraging when examined together with the Real-class F1-scores. The comparatively low Real F1-scores (32%–43%) indicate that all architectures exhibit a high false positive rate by frequently misclassifying authentic images as manipulated. This behavior is likely attributed to the substantial distribution shift between the high-quality authentic images contained in the FF++ benchmark and the comparatively lower-quality images used during training. Consequently, overall accuracy alone provides an overly optimistic assessment of generalization performance. While the FF++ benchmark highlights the challenges associated with significant domain shifts, the DDL benchmark provides a complementary evaluation under image characteristics that more closely resemble the training data.

DDL Evaluation: Compared with FF++, the DDL benchmark demonstrates considerably stronger generalization performance across all evaluated architectures. The models achieved a more balanced trade-off between Fake and Real F1-scores, suggesting that the reduced domain shift enabled more reliable discrimination between authentic and manipulated facial images. Interestingly, EfficientViT emerged as the most robust architecture, achieving the highest overall accuracy (64%) together with the best Real F1-score (56%), thereby outperforming the in-domain leader, MobileViTv2. This finding suggests that although MobileViTv2 provides superior in-domain performance, EfficientViT exhibits stronger robustness when evaluated on previously unseen datasets, highlighting the importance of cross-dataset evaluation when assessing lightweight Vision Transformer architectures for practical digital forensic applications.

VI. CONCLUSION AND FUTURE WORK

The research provides a systematic evaluation of four SOTA lightweight ViTs for the task of deepfake detection. Our conclusion is that MobileViTv2 has the most balanced profile for edge-device forensics. When optimized with inverse class frequency weighting and aggressive regularization (Mixup/CutMix), the architecture achieved a peak in-domain accuracy of 96.3% while maintaining the lowest single-inference latency of 2.56ms and the smallest memory footprint with only 4.39M parameters. Our research approach is restricted to single-face image detection. Additionally, models exhibit a bias towards the central face, ignoring other manipulations. Future models should be training utilizing the forgery masks, to anchor model attention towards the correct artifacts. Additionally, the focus should shift from image samples to videos as they are the more dominant form of media in the current era.

VII. ACKNOWLEDGEMENT

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